

Adversarial Attacks and Defenses on a Sentiment Analysis Model

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Abstract

Machine learning models are increasingly vulnerable to adversarial attacks that subtly modify input data to cause incorrect predictions. This project investigates the robustness of a sentiment analysis classifier trained on the IMDB movie reviews dataset. A baseline model is trained and evaluated using standard performance metrics. Two adversarial attacks—synonym substitution and character-level noise—are applied to evaluate the model’s vulnerability. Subsequently, three defense techniques—adversarial training, regularization, and vocabulary pruning—are implemented to improve robustness. Experimental results demonstrate how adversarial attacks degrade model performance and how defense techniques help recover accuracy and reliability.

1. Introduction

Natural Language Processing (NLP) models are widely used in applications such as sentiment analysis, spam detection, and recommendation systems. Despite their effectiveness, these models are susceptible to adversarial attacks, where small and often imperceptible modifications to text inputs can significantly affect predictions.

This project aims to:

1. Build a machine learning classifier for sentiment analysis.
 2. Evaluate its performance on clean data.
 3. Apply adversarial attacks and observe their impact.
 4. Apply defense techniques to improve robustness.
 5. Compare results before attack, after attack, and after defense.
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2. Dataset Description

The **IMDB Movie Reviews Dataset** is used in this project. It consists of movie reviews labeled as:

- **Positive (1)**
- **Negative (0)**

Dataset Characteristics:

- Binary classification task
 - Balanced dataset
 - Text-based data requiring preprocessing
 - Widely used benchmark for sentiment analysis
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3. Data Preprocessing

The following preprocessing steps were applied:

- Text cleaning (lowercasing and removing unnecessary symbols)
- Tokenization of text data
- Conversion of text into numerical representations using vectorization
- Splitting data into training and testing sets

These steps ensure that the textual data is properly formatted for machine learning model training.

4. Model Development

A supervised machine learning classifier suitable for text classification was selected and trained on the preprocessed dataset.

Training Process:

- Model trained on clean IMDB reviews
 - Hyperparameters tuned for optimal performance
 - Training conducted using standard machine learning practices
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5. Baseline Model Evaluation

The baseline model was evaluated using the following metrics:

- **Accuracy**
- **Precision**
- **Recall**
- **F1-score**
- **Confusion Matrix**

Baseline Results:

The model achieved strong performance on clean test data, demonstrating its ability to correctly classify movie reviews under normal conditions.

6. Adversarial Attacks

Two different adversarial attacks were applied to test the robustness of the trained model.

6.1 Synonym Substitution Attack

Description:

Key words in the review text were replaced with synonyms that preserve semantic meaning but alter the surface form of the text.

Example:

- Original: *"This movie is amazing and enjoyable."*
- Attacked: *"This film is astonishing and pleasant."*

Impact:

Although the meaning remains the same, the model's confidence and accuracy decreased, indicating vulnerability to semantic-level perturbations.

6.2 Character-Level Noise Attack

Description:

Random characters were inserted, removed, or replaced within words.

Example:

- Original: *"The acting was fantastic."*
- Attacked: *"The act1ng was fantast1c."*

Impact:

This attack caused a noticeable drop in accuracy as the model struggled with altered word structures.

7. Performance After Attacks

After applying both attacks, the model showed:

- Reduced accuracy
- Lower precision and recall
- Increased misclassifications

This confirms that the baseline model is vulnerable to adversarial manipulation.

8. Defense Techniques

Three defense techniques were applied to improve model robustness.

8.1 Adversarial Training

The model was retrained using a mixture of clean and adversarially modified samples.

Effect:

- Improved resistance to both attacks
 - Better generalization to perturbed inputs
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8.2 Regularization

Stronger regularization was applied during training to reduce overfitting.

Effect:

- Improved stability
 - Reduced sensitivity to small input changes
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8.3 Vocabulary Pruning

Rare and noisy tokens were removed from the vocabulary.

Effect:

- Reduced impact of character-level noise
 - Improved robustness to spelling variations
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9. Results Comparison

The performance comparison is summarized as follows:

Model State	Accuracy	Precision	Recall	F1-score
Before Attack	High	High	High	High
After Attack	Significantly Lower	Lower	Lower	Lower
After Defense	Improved	Improved	Improved	Improved

The defenses successfully mitigated the negative effects of adversarial attacks.

10. Discussion

The experiments demonstrate that:

- Adversarial attacks can significantly degrade NLP model performance.
 - Different attack types exploit different weaknesses.
 - Combining multiple defense techniques provides better robustness than a single method.
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11. Conclusion

This project successfully demonstrated the full adversarial machine learning workflow:

- A sentiment analysis model was trained and evaluated.
- Two adversarial attacks were applied and analyzed.
- Three defense techniques were implemented to improve robustness.

The results highlight the importance of considering adversarial robustness when deploying NLP models in real-world applications.

12. References

1. IMDB Movie Reviews Dataset
2. Research literature on adversarial attacks in NLP
3. Machine Learning and Natural Language Processing resources